Graph metrics and characterization

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To characterize the structure of the road graph generated from each map and analyze how the different planning approaches behave under varying topologies, we extracted a total of 35 graph metrics per instance. These metrics capture complementary aspects such as size and scale, degree-based connectivity, redundancy and meshing, shortest-path structure, and edge-length statistics. They are used throughout the evaluation to study trends in cost and design decisions as a function of the underlying graph characteristics. Below, the definitions of the computed metrics are given.

Let $G = (V, E)$ be an undirected graph, with $n = |V|$ nodes and $m = |E|$ edges. Each edge $e \in E$ has a positive length $\ell(e) > 0$. Shortest-path distances $d(u, v)$ are computed using $\ell(\cdot)$ as weights [12, 6]. When a metric requires connectivity (e.g., radius/diameter/centralities), it is computed on the largest connected component.

Size and scale

- **Number of nodes (n_nodes):** $n = |V|$. *Counts intersections; problem size.*
- **Number of edges (n_edges):** $m = |E|$. *Counts street or road segments; network size.*
- **Total edge length (total_length):**

$$L_{\text{total}} = \sum_{e \in E} \ell(e). \tag{1}$$

  Total physical extent of the network.

Degree-based connectivity Let $\deg(v)$ denote the degree of node v.

- **Average node degree (avg_degree):**

$$\overline{k} = \frac{1}{n} \sum_{v \in V} \deg(v). \tag{2}$$

  Typical number of streets per intersection. [12]

- **Number of degree-1 nodes (num_deg1):**

$$n_{\deg=1} = |\{v \in V : \deg(v) = 1\}|. \tag{3}$$

  Number of dead ends (leaf nodes).

- **Proportion of degree-1 nodes (prop_deg1):**

$$p_{\deg=1} = \frac{n_{\deg=1}}{n}. \tag{4}$$

  Fraction of dead ends; tree-likeness indicator.

- **Number of nodes with degree ≥ 3 (num_deg_ge3):**

$$n_{\deg \geq 3} = |\{v \in V : \deg(v) \geq 3\}|. \tag{5}$$

Number of branching intersections.

- **Proportion of nodes with degree ≥ 3 (prop_deg_ge3):**

$$p_{\deg \geq 3} = \frac{n_{\deg \geq 3}}{n}. \tag{6}$$

Fraction of branching intersections; junction complexity.

- **Degree entropy (degree_entropy):**

$$H_{\deg} = -\sum_{k} p(k) \log p(k), \tag{7}$$

where $p(k)$ is the empirical probability of degree k. *Heterogeneity of node connectivity.* [12]

Density and mesh structure

- **Graph density (density):**

$$\rho = \frac{2m}{n(n-1)}. \tag{8}$$

Edge density relative to a complete graph [12]. For multigraphs (e.g., graphs with pairs of nodes with multiple parallel streets), the edge count includes parallel edges, so this quantity should be interpreted as an edge-density indicator rather than as the classical simple-graph density.

- **Meshedness coefficient / mesh density (mesh_density):**

$$\rho_{\text{mesh}} = \frac{m - n + 1}{2n - 5}, \qquad (n \geq 3). \tag{9}$$

Planar-oriented meshedness (cyclic redundancy normalized by a planar upper bound), useful for street networks. [2]

Shortest-path extension metrics Eccentricity is defined as:

$$\epsilon(u) = \max_{v \in V} d(u, v). \tag{10}$$

- **Average shortest path length (avg_shortest_path_length):**

$$\overline{d} = \frac{1}{n(n-1)} \sum_{u \neq v} d(u, v). \tag{11}$$

Typical travel distance within the network. [12]

- **Graph radius (radius):**

$$r = \min_{u \in V} \epsilon(u). \tag{12}$$

Best achievable "center" (minimizes the farthest distance). [6]

- **Graph diameter (diameter):**

$$D = \max_{u \in V} \epsilon(u). \tag{13}$$

Maximum shortest-path distance; spatial extent. [6]

- **Average eccentricity (avg_eccentricity):**

$$\bar{\epsilon} = \frac{1}{n} \sum_{u \in V} \epsilon(u). \tag{14}$$

Average remoteness of nodes.

Centrality metrics

- **Average closeness centrality (avg_node_closeness):** Closeness centrality measures how close a node is to all other nodes in terms of shortest-path distance. It is defined as:

$$CC(u) = \frac{n-1}{\sum_{v \neq u} d(u,v)}, \tag{15}$$

where $d(u, v)$ is the shortest-path distance between nodes u and v. In this work, distances are computed using edge lengths, and closeness centrality was computed using the NetworkX implementation [4].

The average closeness centrality is then:

$$\overline{CC} = \frac{1}{n} \sum_{u \in V} CC(u). \tag{16}$$

Higher values indicate nodes that are, on average, closer to all others in the network [8]

- **Maximum betweenness centrality (max_node_betweenness):** Let σ_{st} be the number of shortest paths from node s to node t, and $\sigma_{st}(u)$ the number of those paths that pass through node u. Betweenness centrality is defined as:

$$BC(u) = \sum_{s \neq u \neq t} \frac{\sigma_{st}(u)}{\sigma_{st}}. \tag{17}$$

In this work, shortest paths are computed using edge lengths as weights. The values are normalized following the standard convention so that $BC(u) \in [0, 1]$. The maximum betweenness centrality is:

$$BC_{\max} = \max_{u \in V} BC(u). \tag{18}$$

Nodes with high betweenness act as bottlenecks for shortest-path flow [7, 1]

- **Average betweenness centrality (avg_node_betweenness):**

$$\overline{BC} = \frac{1}{n} \sum_{u \in V} BC(u). \tag{19}$$

Represents the typical importance of nodes for shortest-path flow.

- **Coefficient of variation of betweenness (cv_node_betweenness):**

$$CV_{BC} = \frac{\mathrm{std}(\{BC(u)\})}{\mathrm{mean}(\{BC(u)\})}, \tag{20}$$

defined when the mean is nonzero. *Measures how unevenly centrality is distributed across nodes.*

Branching, cycles, and spanning structure

- **Branching index (branching_index):**

$$B = \frac{n_{\deg=1}}{n_{\deg\geq 3}}. \tag{21}$$

Tree-likeness proxy: many dead ends vs few branching junctions.

- **Cyclomatic number / circuit rank (cyclomatic_number):**

$$\mu = m - n + c, \tag{22}$$

where c is the number of connected components. *Number of "extra" edges beyond a spanning forest; counts independent cycles* [9, 6]

- **Minimum spanning tree total length (mst_length):** Let T be a minimum spanning tree (MST) using $\ell(\cdot)$.

$$L_{\mathrm{MST}} = \sum_{e\in T} \ell(e). \tag{23}$$

Minimum total length to connect all nodes (tree backbone) [11, 15]

- **MST redundancy ratio (mst_ratio):**

$$R_{\mathrm{MST}} = \frac{L_{\mathrm{total}}}{L_{\mathrm{MST}}}, \tag{24}$$

defined when $L_{\mathrm{MST}} > 0$. *Redundancy vs a tree: how much longer the real network is than its MST.*

Edge-length distribution statistics Let $\mathcal{L} = \{\ell(e) : e \in E\}$.

- **Mean edge length (avg_edge_length):**

$$\mu_\ell = \mathrm{mean}(\mathcal{L}). \tag{25}$$

Typical segment length.

- **Median edge length (median_edge_length):**

$$\tilde{\ell} = \mathrm{median}(\mathcal{L}). \tag{26}$$

Robust typical length.

- **Standard deviation of edge length (std_edge_length):**

$$\sigma_\ell = \mathrm{std}(\mathcal{L}). \tag{27}$$

Absolute dispersion of segment lengths.

- **Coefficient of variation of edge length (cv_edge_length):**

$$CV_\ell = \frac{\sigma_\ell}{\mu_\ell}, \tag{28}$$

defined when $\mu_\ell > 0$. *Relative heterogeneity of segment lengths (scale-free)* [10]

- **Minimum edge length (min_edge_length):**

$$\ell_{\min} = \min(\mathcal{L}). \tag{29}$$

Shortest segment present.

- **Maximum edge length (max_edge_length):**

$$\ell_{\max} = \max(\mathcal{L}). \tag{30}$$

Longest segment present.

- **25th percentile of edge length (edge_length_p25):**

$$\ell_{25} = \text{quantile}_{0.25}(\mathcal{L}). \tag{31}$$

Lower-typical segment length.

- **75th percentile of edge length (edge_length_p75):**

$$\ell_{75} = \text{quantile}_{0.75}(\mathcal{L}). \tag{32}$$

Upper-typical segment length.

Community structure.

- Modularity (modularity). A community is defined as a group of nodes that are more densely connected among themselves than with the rest of the network. In this work, communities are identified using the `greedy_modularity_communities` function from NetworkX [5], which uses Clauset-Newman-Moore greedy modularity maximization [3] to approximate the community partition with the largest modularity. Modularity is defined following [13], and its value is computed using the NetworkX implementation, which follows the standard formulation:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \tag{33}$$

where A_{ij} is the adjacency matrix, k_i and k_j are node degrees, and $\delta(c_i, c_j)$ equals 1 if nodes i and j belong to the same community, and 0 otherwise. *Higher values indicate stronger community structure.*

- Number of communities (n_communities). The number of communities corresponds to the number of groups returned by the `greedy_modularity_communities` function in NetworkX. *It indicates how many distinct modules the network is partitioned into under this optimization criterion.*

Average straightness (avg_straightness_length). For a pair of nodes (u, v), straightness is defined as the ratio between the Euclidean distance $d_E(u, v)$ and the network shortest-path distance $d_G(u, v)$, following spatial network analysis approaches:

$$S(u, v) = \frac{d_E(u, v)}{d_G(u, v)}. \tag{34}$$

Let P denote the set of node pairs considered in the computation, and $|P|$ its cardinality. We define the graph-level straightness as the average of this quantity over P:

$$\overline{S} = \frac{1}{|P|} \sum_{(u,v) \in P} \frac{d_E(u, v)}{d_G(u, v)}. \tag{35}$$

This metric is inspired by the definition of straightness centrality [14], and quantifies how direct routes are compared to straight-line distances.

Average tortuosity (avg_tortuosity_length). Tortuosity is defined as the inverse of straightness for each node pair:

$$T(u,v) = \frac{d_G(u,v)}{d_E(u,v)}. \tag{36}$$

Using the same set of node pairs P, the graph-level average tortuosity is defined as:

$$\overline{T} = \frac{1}{|P|} \sum_{(u,v)\in P} \frac{d_G(u,v)}{d_E(u,v)}. \tag{37}$$

This metric measures the detour imposed by the network relative to the Euclidean distance.

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